Math 4261: Mathematical Statistics I

Lecture 18: Review 2

Fall 2026

Outline

MLE

Bayesian inference

Model misspecification

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Model misspecification

Parameter estimation and MLE

Suppose data $X_1, \ldots, X_n$ came from joint distribution with PMF/PDF parametrized by $\theta \in \Omega \subseteq \mathbb{R}^k$:

$$f(x_1, \ldots, x_n \mid \theta).$$

Goal: estimate θ.

An **estimator** $\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n)$ is a function of the data. $\widehat{\theta}$ is a random variable.

The **likelihood function** and **log-likelihood** are

$$\text{lik}(\theta) = f(X_1, \ldots, X_n \mid \theta), \qquad \ell_n(\theta) = \log \text{lik}(\theta).$$

The **maximum likelihood estimator** is

$$\widehat{\theta} = \operatorname*{argmax}_{\theta \in \Omega} \ell_n(\theta).$$

"MLE $\widehat{\theta}$ is the value of θ that maximizes the joint PMF/PDF at the observed data".

Asymptotic normality

Consider an estimator $\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n)$ of θ. Here $X_1, \ldots, X_n$ are data from some distribution parametrized by θ.

- Asymptotic normality means

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta)).$$

- In particular, $\widehat{\theta} \xrightarrow{p} \theta$: consistency.
- Large-sample approximation: $\widehat{\theta} \sim \mathcal{N}\left(\theta, \frac{v(\theta)}{n}\right)$.
- Approximate standard error is $\sqrt{\frac{v(\theta)}{n}}$,

  with plug-in estimate $\sqrt{\frac{v(\widehat{\theta})}{n}}$

Main MLE theorem

Theorem

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x|\theta)$, where $\theta \in \Omega \subseteq \mathbb{R}$ is a single parameter. Let $\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n)$ denote the MLE. Under regularity conditions on $f(x \mid \theta)$,

- $\widehat{\theta}$ is consistent: $\widehat{\theta} \to \theta$ in probability.
- $\widehat{\theta}$ is asymptotically normal: $\sqrt{n}\left(\widehat{\theta} - \theta\right) \xrightarrow{d} \mathcal{N}(0, \frac{1}{I(\theta)})$.

Here $I(\theta)$ is the single observation **Fisher information** for $X \sim f(x \mid \theta)$:

$$I(\theta) = \text{Var}_\theta\left[\frac{\partial}{\partial\theta} \log f(X \mid \theta)\right] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \log f(X \mid \theta)\right],$$

where $\mathbb{E}_\theta$, Var_θ denote expectation/variance over $X \sim f(x \mid \theta)$.

Fisher information

Main insight from MLE theorem:

$$\boxed{\widehat{\theta} \sim \mathcal{N}\left(\theta, \frac{1}{nI(\theta)}\right), \quad \text{approximately}}$$

Here $nI(\theta)$ is the **Fisher information in the sample**, which is more generally

$$I_n(\theta) = \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\ell_n(\theta)\right] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\ell_n(\theta)\right]$$

In **i.i.d.** setting, we have $I_n(\theta) = nI(\theta)$, where $I(\theta)$ is the **Fisher information in a single observation**.

Cramer-Rao lower bound

Theorem

Under regularity conditions, let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ where θ is a single parameter. Let $T = T(X_1, \ldots, X_n)$ be any unbiased estimator of θ. Then

$$\text{Var}_\theta[T] \geq \frac{1}{nI(\theta)}.$$

- No unbiased estimator can have variance below this bound, regardless of how much computation or ingenuity we use in creating it.
- The **MLE** attains this bound!

Asymptotic efficiency

An estimator $\widehat{\theta}$ is **asymptotically efficient** if

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta)}\right).$$

That is, the variance of the Normal approximation to $\widehat{\theta}$ for large n attains the Cramer-Rao lower bound of $\frac{1}{nI(\theta)}$. Our main theorem thus shows that the MLE is asymptotically efficient.

If two estimators $\widehat{\theta}_1$ and $\widehat{\theta}_2$ based on $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ satisfy

$$\sqrt{n}(\widehat{\theta}_1 - \theta) \xrightarrow{d} \mathcal{N}(0, v_1(\theta))$$
$$\sqrt{n}(\widehat{\theta}_2 - \theta) \xrightarrow{d} \mathcal{N}(0, v_2(\theta))$$

where $v_2(\theta) < v_1(\theta)$, then $v_2(\theta)/v_1(\theta)$ is the **asymptotic relative efficiency** of $\widehat{\theta}_1$ relative to $\widehat{\theta}_2$.

Fisher information for multiple parameters

For a parametric model with p parameters $\theta \in \Omega \subseteq \mathbb{R}^p$, the **Fisher information matrix** $I(\theta) \in \mathbb{R}^{p \times p}$ is the matrix whose (i, j) entry is

$$
\begin{aligned}
I(\theta)_{i,j} &= \operatorname{Cov}_\theta \left[\frac{\partial}{\partial \theta_i} \log f(X \mid \theta), \frac{\partial}{\partial \theta_j} \log f(X \mid \theta) \right] \\
&= -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(X \mid \theta) \right].
\end{aligned}
$$

Theorem

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$, where $\theta \in \Omega \subseteq \mathbb{R}^p$ is the parameter vector. Under regularity assumptions

$$
\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}).
$$

That is, the limiting distribution is a p-dimensional multivariate Gaussian whose covariance is the $p \times p$ matrix inverse $I(\theta)^{-1}$.

Confidence intervals

An interval $\widehat{I} = \widehat{I}(X_1, \ldots, X_n)$ is an asymptotic $(1-\alpha)$-confidence interval for θ if

$$\mathbb{P}_\theta[\theta \in \widehat{I}] \longrightarrow 1-\alpha \qquad \text{as } n \to \infty.$$

Suppose $\widehat{\theta}$ is consistent and asymptotically normal:

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta)),$$

where v is continuous and positive at θ. Then we can use

$$\widehat{I} = \left[\widehat{\theta} - z^{(\alpha/2)} \sqrt{\frac{v(\widehat{\theta})}{n}}, \quad \widehat{\theta} + z^{(\alpha/2)} \sqrt{\frac{v(\widehat{\theta})}{n}} \right],$$

where $z^{(\alpha/2)}$ is the upper-$\alpha/2$ point of the standard Normal.

When using the MLE $\widehat{\theta}$, put $v(\theta) = 1/I(\theta)$.

Example: MLE (Review 2 Q1)

Assume the alleles S and s are present in a population with proportions θ and $1-\theta$, respectively. Hardy–Weinberg equilibrium models each of the two alleles a person inherits as an independent trial with either an S or s outcome with probability θ and $1-\theta$ respectively. Individuals' genotypes are independent.

Let Y_1, Y_2, Y_3 count the individuals with SS, Ss/sS, and ss genotypes in a population of size n. The likelihood is

$$\text{lik}(\theta) = \frac{n!}{Y_1!Y_2!Y_3!} p_1^{Y_1} p_2^{Y_2} p_3^{Y_3}, \qquad \theta \in [0,1].$$

Write down p_1, p_2, p_3 in terms of θ.

$p_1 = \theta^2$

$p_2 = 2\theta(1-\theta)$

$p_3 = (1-\theta)^2$

Example: MLE (Review 2 Q1)

Find the MLE $\hat{\theta}$.

does not involve θ

$$\ell_n(\theta) = \text{Const} + Y_1 \log \theta^2 + Y_2 \log(2\theta(1-\theta)) + Y_3 \log(1-\theta)^2$$

$$= \text{Const.} + 2Y_1 \log\theta + Y_2 \log\theta + Y_2 \log(1-\theta) + 2Y_3 \log(1-\theta)$$

$$= \text{Const.} + (2Y_1 + Y_2)\log\theta + (Y_2 + 2Y_3)\log(1-\theta)$$

$$\ell_n'(\theta) = \frac{2Y_1 + Y_2}{\theta} - \frac{Y_2 + 2Y_3}{1-\theta} \overset{\text{Set}}{=} 0$$

Solving: $(1-\theta)(2Y_1 + Y_2) - \theta(Y_2 + 2Y_3) = 0$

$$-\theta(2Y_1 + 2Y_2 + 2Y_3) = -(2Y_1 + Y_2)$$

$2n$

$$\boxed{\hat{\theta} = \frac{2Y_1 + Y_2}{2n}}$$

Example: MLE (Review 2 Q1)

Show that the Fisher information in the whole sample is

$$I_n(\theta) = \frac{2n}{\theta(1-\theta)}, \qquad 0 < \theta < 1.$$

$$I_n(\theta) = -\mathbb{E}_\theta[\ell_n''(\theta)] = -\mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\left(\frac{2Y_1+Y_2}{\theta} - \frac{Y_2+2Y_3}{1-\theta}\right)\right]$$

$$= -\mathbb{E}_\theta\left[-\frac{2Y_1+Y_2}{\theta^2} - \frac{Y_2+2Y_3}{(1-\theta)^2}\right]$$

$Y_j \sim \text{Bin}(n, p_j)$

$\mathbb{E}\, Y_j = n p_j$

$$= \frac{2\mathbb{E}_\theta[Y_1] + \mathbb{E}_\theta Y_2}{\theta^2} + \frac{\mathbb{E}_\theta Y_2 + 2\mathbb{E}_\theta Y_3}{(1-\theta)^2}$$

p_1

$$= \frac{2n\cdot\theta^2 + n2\theta(1-\theta)}{\theta^2} + \frac{n2\theta(1-\theta) + 2n(1-\theta)^2}{(1-\theta)^2}$$

Example: MLE (Review 2 Q1)

Show that the Fisher information in the whole sample is

$$I_n(\theta) = \frac{2n}{\theta(1-\theta)}, \qquad 0 < \theta < 1.$$

Cont'd

$$I_n(\theta) = \cdots = \frac{2n\cdot\theta^2 + n2\theta(1-\theta)}{\theta^2} + \frac{n2\theta(1-\theta) + 2n(1-\theta)^2}{(1-\theta)^2}$$

$$= \frac{2n}{\theta} + \frac{2n}{1-\theta}$$

$$= \frac{2n}{\theta(1-\theta)}$$

Example: MLE (Review 2 Q1)

Blood samples were collected for a population of size 500:

Genotype	SS	Ss/sS	ss	Total
Count	325	150	25	500

Construct an approximate 95%-confidence interval for θ. You may use $z^{(0.025)} = 1.96$.

This does not fit in our usual iid MLE setting, but we can assume

Approximately $\hat{\theta} \sim N(\theta, \frac{1}{I_n(\theta)}) = N(\theta, \frac{\theta(1-\theta)}{2n})$

95%-CI $\left[\hat{\theta} - 1.96\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{2n}}, \hat{\theta} + 1.96\sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{2n}} \right]$

where $\hat{\theta} = \frac{2Y_1 + Y_2}{2n} = \frac{2(325) + 150}{1000} = \frac{8}{10}$

Delta method

Motivation: Suppose we estimate θ for $X_1, \ldots, X_n \sim \text{Pareto}(\theta, 1)$ with $\widehat{\theta}$, say with MLE or MoM. The mean of this distribution is $\mathbb{E}X = \frac{\theta}{\theta - 1} =: g(\theta)$. The **plug-in** estimate is $g(\widehat{\theta})$.

The **delta method** tells us how to transfer asymptotic normality properties for $\widehat{\theta}$ to $g(\widehat{\theta})$.

Theorem (Delta method)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ and uppose $S = S(X_1, \ldots, X_n)$ satisfies

$$\sqrt{n}(S - \mu(\theta)) \xrightarrow{d} \mathcal{N}(0, v(\theta))$$

If $g : I \to \mathbb{R}$ is continuously differentiable at $\mu(\theta)$, where I is an open interval containing $\mu(\theta)$, then

$$\sqrt{n}(g(S) - g(\mu(\theta))) \xrightarrow{d} \mathcal{N}(0, g'(\mu(\theta))^2 v(\theta)).$$

Example: Delta method (Review 2 Q2)

Let $X_1, X_2, \ldots, X_n, \ldots$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ random variables, where $\sigma^2 > 0$. Note that $\mathbb{E}[X_i^4] = 3\sigma^4$.

Consider $Y_n := \frac{1}{n}\sum_{i=1}^n X_i^2$. Use the delta method to find constants a_n and b so that

$\mathbb{E}X^2 = \sigma^2$, $\mathrm{Var}\,X^2 = \mathbb{E}X^4 - (\mathbb{E}X^2)^2$
$= 3\sigma^4 - \sigma^4$
$= 2\sigma^4$

$$a_n(1/Y_n - b) \xrightarrow{d} \mathcal{N}(0,1) \qquad \text{as } n \to \infty.$$

CLT gives $\sqrt{n}(Y_n - \sigma^2) \xrightarrow{d} \mathcal{N}(0, 2\sigma^4)$

Define $g(t) = \frac{1}{t}$ which is continuous on $t > 0$.

$g'(t) = -\frac{1}{t^2}$, $g'(\sigma^2) = -\frac{1}{\sigma^4}$

Delta method $\sqrt{n}\left(\frac{1}{Y_n} - \frac{1}{\sigma^2}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{\sigma^8} \cdot 2\sigma^4\right)$ $= \frac{2}{\sigma^4}$

i.e. $\sqrt{n}\,\frac{\sigma^2}{\sqrt{2}}\left(\frac{1}{Y_n} - \frac{1}{\sigma^2}\right) \xrightarrow{d} \mathcal{N}(0,1)$

Outline

MLE

Bayesian inference

Model misspecification

Bayesian approach to parameter estimation

Given data $\boldsymbol{X} \sim f(\boldsymbol{x} \mid \theta)$ from a parametric model, Bayesian inference models θ as random, with a prior distribution

$$\Theta \sim f_{\Theta}(\theta)$$

Bayesian inference centers around the posterior distribution

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x})$$

representing our updated beliefs about Θ, after ('post') observing the data $\boldsymbol{X} = \boldsymbol{x}$.

The posterior may be computed from

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x}) \propto f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) \cdot f_{\Theta}(\theta)$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

Point estimate

After obtaining the posterior distribution, we typically report the **posterior mean**

$$\delta_{\text{Bayes}}(\boldsymbol{X}) = \mathbb{E}[\Theta \mid \boldsymbol{X}].$$

as a point estimate of Θ.

In fact, in terms of minimizing the MSE

$$\text{MSE}(\delta) = \mathbb{E}[(\Theta - \delta(\boldsymbol{X}))^2].$$

over all possible estimators δ, the posterior mean is the **best estimator**!

Credible sets

A Bayesian credible interval $I(\boldsymbol{X})$ with coverage $1-\alpha$ is an interval which guarantees

$$\mathbb{P}[\Theta \in I(\boldsymbol{X}) \mid \boldsymbol{X} = \boldsymbol{x}] = 1-\alpha.$$

Here Θ is random, and $1-\alpha$ is the probability of $\Theta \in I(\boldsymbol{x})$ over the randomness of Θ, given the observed data $\boldsymbol{X} = \boldsymbol{x}$.

Example: In Homework 6 Q1, we found the Gaussian posterior

$$\Theta \mid \boldsymbol{X} \sim \mathcal{N}\left(\mu_{\text{post}}, \frac{1}{\xi_{\text{post}}}\right).$$

A 95% credible set is

$$\left[\mu_{\text{post}} - \frac{1.96}{\sqrt{\xi_{\text{post}}}}, \quad \mu_{\text{post}} + \frac{1.96}{\sqrt{\xi_{\text{post}}}}\right].$$

Useful distributions

$Y \sim \text{Beta}(\alpha, \beta)$ if Y is continuous with PDF

$$f_Y(y) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1}(1-y)^{\beta-1}, \quad \text{for } y \in (0,1)$$

where B is the Beta function. $\mathbb{E}[Y] = \frac{\alpha}{\alpha+\beta}$.

$Y \sim \text{Gamma}(\alpha, \beta)$ if Y is continuous with PDF

$$f_Y(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, \quad \text{for } y > 0$$

where Γ is the Gamma function. $\mathbb{E}[Y] = \frac{\alpha}{\beta}$.

Example: Beta–Bernoulli model

Let $X_1, \ldots, X_n \sim \mathsf{Bern}(p)$ i.i.d. with prior $P \sim \mathrm{Beta}(\alpha, \beta)$. We found that the posterior distribution is

$$P \mid \boldsymbol{X} \sim \mathrm{Beta}(S + \alpha, n - S + \beta), \qquad S = X_1 + \cdots + X_n.$$

The posterior mean is

$$\widehat{p} = \mathbb{E}\left[P \mid \boldsymbol{X}\right] = \frac{S + \alpha}{n + \alpha + \beta}$$

This is different from the MLE/MoM estimator $\overline{X} = S/n$. It is biased towards the prior mean.

Example: Posterior (Review 2 Q4)

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}(\mu, 1/\xi)$ where we parametrize by $\xi = 1/\sigma^2$. Suppose μ is known, and ξ is unknown and to be estimated. Model ξ as a random variable with prior distribution $\Xi \sim \text{Gamma}(\alpha, \beta)$, with $\alpha, \beta > 0$.

Compute the posterior and state the posterior mean.

Likelihood $f_{X|\Xi}(x_1, \ldots, x_n | \xi) = \prod_{i=1}^{n} \sqrt{\frac{\xi}{2\pi}} \exp\left\{-\frac{\xi}{2}(x_i - \mu)^2\right\}$

Prior $f_{\Xi}(\xi) = \frac{\beta^\alpha}{\Gamma(\alpha)} \xi^{\alpha-1} e^{-\beta\xi}$

Posterior

$$f_{\Xi|X}(\xi | x_1 \ldots x_n) \propto f_{X|\Xi}(x_1 \ldots x_n | \xi) f_{\Xi}(\xi)$$

$$\propto \xi^{n/2} e^{-\xi \frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^2} \cdot \xi^{\alpha-1} e^{-\xi\beta}$$

Example: Posterior (Review 2 Q4)

$$f_{\Xi|X}(\xi|x_1 \dots x_n) \propto \xi^{\frac{n}{2}+\alpha-1} e^{-\xi\left(\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2+\beta\right)}$$

Posterior

$$\Xi|X \sim \text{Gamma}\left(\frac{n}{2}+\alpha, \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2+\beta\right)$$

Example: Bayesian application

Suppose we wish to determine the accuracy of a measuring instrument. The measurement errors are known to follow a $\mathcal{N}(0, 1/\xi)$ distribution. We model the unknown precision as a random variable Ξ. Prior experience informs us that $\Xi \sim \text{Gamma}(3, 6)$.

We conduct 6 measurements of a rod known to be 100 cm in length:

101	99.5	100	102	98	100.5

Compute the posterior distribution and use the posterior mean to estimate the precision ξ of the measurement errors.

Errors $\underset{x_1}{1}$ $\underset{x_2}{-\frac{1}{2}}$ $\underset{x_3}{0}$ $\underset{x_4}{2}$ $\underset{x_5}{-2}$ $\underset{x_6}{\frac{1}{2}}$ $X_i \overset{iid}{\sim} N(0, \frac{1}{\xi})$

Posterior $\Xi \mid X = x \sim \text{Gamma}\left(3+3, \underbrace{\frac{1}{2}\sum_{i=1}^{6} x_i^2 + 6}_{43/4}\right)$

Example: Bayesian application

$$\sum_{i=1}^{6} x_i^2 = 1^2 + \left(\tfrac{1}{2}\right)^2 + 0^2 + 2^2 + (-2)^2 + \left(\tfrac{1}{2}\right)^2$$

$$= 19/2$$

Posterior mean $\quad E[\Omega \mid X = x] = \dfrac{6}{63/4} = 24/63$.

Outline

MLE

Bayesian inference

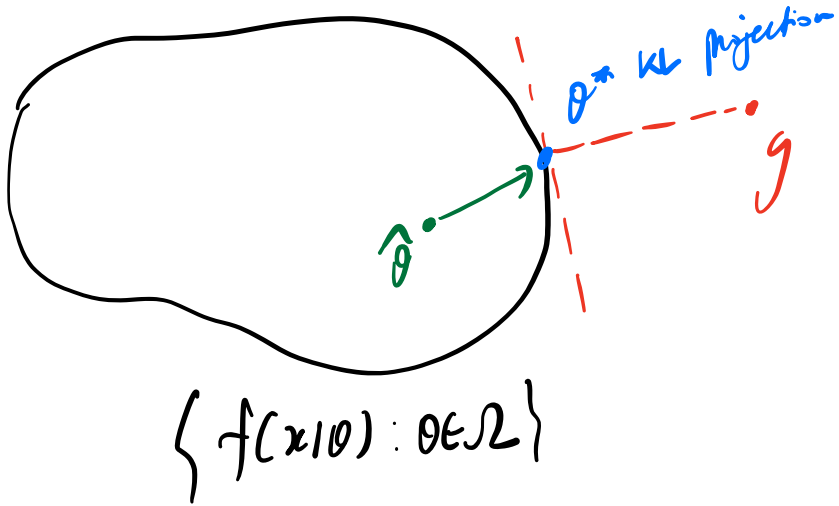
Model misspecification

Setting

Suppose data $X_1, \ldots, X_n \sim g$ i.i.d. for some distribution g.

We fit a model $\{f(x \mid \theta) : \theta \in \Omega\}$. Suppose we use the MLE procedure to obtain $\widehat{\theta}$.

We aim to give a **theoretical interpretation** of $\widehat{\theta}$.



Main theorem on misspecified models

Theorem

Let $D_{\mathrm{KL}}(g(x) \,\|\, f(x \mid \theta))$ be the Kullback–Leibler (KL) divergence of $g(x)$ relative to $f(x \mid \theta)$, and suppose that

$$\theta \mapsto D_{\mathrm{KL}}(g(x) \,\|\, f(x \mid \theta))$$

has a **unique minimizer** $\theta^* \in \Omega$. Then, under regularity conditions for $f(x \mid \theta)$ and g, the MLE $\widehat{\theta}$ converges to θ^* in probability as $n \to \infty$.

That is, for large n, the MLE estimates the "KL projection" of the true distribution $g(x)$ onto our model $\{f(x \mid \theta) : \theta \in \Omega\}$.

Kullback–Leibler (KL) divergence

The **Kullback–Leibler (KL) divergence** of distributions g relative to f is

$$\boxed{D_{\mathrm{KL}}(g \,\|\, f) = \mathbb{E}_g\left[\log \frac{g(X)}{f(X)}\right],}$$

where $\mathbb{E}_g$ indicates expectation with respect to $X \sim g(x)$.

- $D_{\mathrm{KL}}(g \,\|\, f) \geq 0$ for any f and g.
- In general, it is not symmetric: $D_{\mathrm{KL}}(f \,\|\, g) \neq D_{\mathrm{KL}}(g \,\|\, f)$.
- Within the same parametric model, for parameters θ_0 and θ_1 close to each other, KL behaves like the squared distance:

$$D_{\mathrm{KL}}(f(x \mid \theta_0) \,\|\, f(x \mid \theta_1)) \approx \frac{I(\theta_0)}{2}(\theta_1 - \theta_0)^2.$$

Asymptotic variance for misspecified MLE

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} g(x)$. Fit a model $f(x \mid \theta)$ over parameters $\theta \in \Omega$ using maximum likelihood.
Define the score and its derivative

$$z(x, \theta) = \frac{\partial}{\partial \theta} \log f(x \mid \theta) \qquad \text{and} \qquad z'(X, \theta) = \frac{\partial}{\partial \theta} z(X, \theta).$$

Theorem (Sandwich theorem)

Under suitable regularity conditions, let $\widehat{\theta}$ and θ^* denote the MLE and its KL-projection, where θ^* is an interior point of $\Omega \subseteq \mathbb{R}$. Then

$$\sqrt{n}\,(\widehat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\operatorname{Var}_g\left[z(X, \theta^*)\right]}{\left(\mathbb{E}_g\left[z'(X, \theta^*)\right]\right)^2}\right).$$

Here Var_g and $\mathbb{E}_g$ are with respect to $X \sim g$.

Example: MLE misspecification (Review 2 Q6)

Suppose that we fit a Poisson(λ) model to i.i.d. data $X_1, \ldots, X_n$. However, the true distribution of the data is geometric:

$$g(x) = \frac{1}{3}\left(\frac{2}{3}\right)^x, \qquad x = 0, 1, 2, \ldots.$$

For this distribution, $\mathbb{E}_g[X] = 2$ and $\mathrm{Var}_g[X] = 6$. The fitted PMF is

$$f(x \mid \lambda) = e^{-\lambda}\frac{\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots, \quad \lambda > 0.$$

(i) Compute $D_{\mathrm{KL}}(g \,\|\, f_\lambda)$ up to an additive constant independent of λ. Find its minimizer $\lambda^\star$.

$$D_{KL}(g \,\|\, f_\lambda) = \mathbb{E}_g\left[\log \frac{g(X)}{f(X|\lambda)}\right] = \mathbb{E}_g\left[\log \frac{\frac{1}{3}\left(\frac{2}{3}\right)^X}{e^{-\lambda}\frac{\lambda^X}{X!}}\right]$$

Example: MLE misspecification (Review 2 Q6)

$$D_{KL}(g \| f_\lambda) = \text{Const} - \mathbb{E}_g \log e^{-\lambda} - \mathbb{E}_g \log \lambda^X$$

$$= \text{Const} + \lambda - (\log \lambda)\mathbb{E}_g X.$$

$$\frac{d}{d\lambda} D_{KL}(g \| f_\lambda) = 1 - \frac{\mathbb{E}_g X}{\lambda} \overset{\text{set}}{=} 0$$

Minimizer $\lambda^* = \mathbb{E}_g X$.

Indeed $\frac{d^2}{d\lambda^2} D_{KL}(g \| f_\lambda) = \frac{\mathbb{E}_g X}{\lambda^2} > 0$

Example: MLE misspecification (Review 2 Q6)

The fitted Poisson MLE is $\widehat{\lambda} = \bar{X}$. We know that $\widehat{\lambda} \xrightarrow{p} \lambda^\star = 2$.

(ii) Apply the Sandwich theorem and find V in

$$\sqrt{n}(\widehat{\lambda} - \lambda^\star) \xrightarrow{d} \mathcal{N}(0, V).$$

Score $z(X,\lambda) = \frac{\partial}{\partial\lambda} \log f(X|\lambda) = \frac{\partial}{\partial\lambda} \log\left[e^{-\lambda}\frac{\lambda^X}{X!}\right]$

$= \frac{\partial}{\partial\lambda}\left[-\lambda + X\log\lambda - \log X!\right]$

$= -1 + \frac{X}{\lambda}$

$z'(X,\lambda) = \frac{d}{d\lambda} z(X,\lambda) = -\frac{X}{\lambda^2}$

Example: MLE misspecification (Review 2 Q6)

The fitted Poisson MLE is $\widehat{\lambda} = \bar{X}$. We know that $\widehat{\lambda} \xrightarrow{p} \lambda^\star = 2$.

(ii) Apply the Sandwich theorem and find V in

$$\sqrt{n}(\widehat{\lambda} - \lambda^\star) \xrightarrow{d} \mathcal{N}(0, V).$$

$$\mathrm{Var}_g\, z(X, \lambda^*) = \mathrm{Var}_g\left[-1 + \frac{X}{2}\right] = \frac{1}{4}\mathrm{Var}_g X = \frac{6}{4} = \frac{3}{2}$$

$$\mathbb{E}_g\, z'(X, \lambda^*) = \mathbb{E}_g\left[-\frac{X}{4}\right] = -\frac{1}{2}$$

Sandwich theorem

$$\sqrt{n}\left(\widehat{\lambda} - \lambda^*\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\frac{3}{2}}{(-\frac{1}{2})^2}\right) = \mathcal{N}(0, 6)$$

Example: MLE misspecification (Review 2 Q6)

(iii) Suppose we incorrectly assumed that the Poisson model was correctly specified. By usual MLE theory, we would estimate the variance of $\widehat{\lambda}$ by

$$\frac{1}{nI(\widehat{\lambda})},$$

where $I(\lambda)$ is the Fisher information in one Poisson observation.

Approximate this quantity for large n. Does it agree with the variance V/n obtained under the true geometric distribution?

$I(\lambda) = \mathrm{Var}_\lambda[z(X,\lambda)] = \mathrm{Var}_\lambda\left[-1 + \frac{X}{\lambda}\right] = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$

For large n, $I(\hat{\lambda}) = I(\bar{X}) \approx I(2) = \frac{1}{2}$

since $\bar{X} \xrightarrow{p} \mathbb{E}_2 X = 2$ by LLN.

This says variance of $\hat{\lambda}$ is $\approx \frac{2}{n} \neq \frac{6}{n} = \frac{V}{n}$.

Example: MLE misspecification (Review 2 Q6)